

Every Patch in the Dark

A complete, step-by-step history of the dark-matter model — from the first inference of unseen mass to the cosmic dipole, and what comes next

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A complete, step-by-step history of the dark-matter model — from the first inference of unseen mass to the cosmic dipole anomaly, and the adjustment to come

How to read this

What follows is a chronology. Each numbered entry is a discrete step in the life of one idea: that the gravity we measure exceeds the matter we can see. I have tried to mark every place where the model *changed* — where a piece was added, a candidate swapped, an assumption weakened — because the shape of that sequence is the real subject. Where the change was a genuine, confirmed prediction, I say so plainly; where it was an after-the-fact accommodation, I say that too. The honest verdict is that the history contains both, and that the ratio has been shifting.

PART I — Two templates (1846–1859)

1. 1846 — Neptune: unseen mass, and it was really there. Uranus wandered off its predicted orbit. Urbain Le Verrier assumed an unseen planet was tugging it, calculated where that planet must be, and within hours of receiving the prediction Johann Galle pointed a telescope and found Neptune almost exactly there. This is the founding success of the “invisible mass” method: a discrepancy in motion, explained by matter we had not yet seen. It is the template dark matter still invokes.

2. 1859 — Vulcan: unseen mass, and it was *not* there. Le Verrier tried the identical trick on Mercury, whose orbit also misbehaved. He predicted a planet, “Vulcan,” inside Mercury’s orbit. Astronomers searched for half a century and found nothing — because the anomaly was not unseen matter at all. It was a flaw in the law of gravity itself, resolved only in 1915 when Einstein’s general relativity explained Mercury’s orbit with no new planet. This is the founding *failure* of the method, and the permanent warning that sits beside it: when motion misbehaves, the cause may be missing matter, or it may be missing physics. Every chapter that follows is a wager on which of these two 19th-century templates the galaxies are repeating.

PART II — The prehistory of missing mass (1884–1932)

3. 1884 — Lord Kelvin counts dark bodies. In his Baltimore Lectures, Kelvin used the velocities of stars in the Milky Way to estimate its mass, and noted that “many of our stars, perhaps a great majority of them, may be dark bodies.” This is the first quantitative argument that the galaxy holds more mass than its luminous stars reveal.

4. 1906 — Poincaré names it. Discussing Kelvin’s estimate, Henri Poincaré wrote of *matière obscure* — “dark matter” — and concluded, reassuringly for the time, that the dark bodies could not greatly outnumber the bright ones. The *term* is now a century old; the *concept*, that gravity outruns light, older still.

5. 1922 — Kapteyn makes the first measurement. Jacobus Kapteyn modeled the motions of stars perpendicular to the galactic plane and produced the first numerical *estimate* of the density of any unseen matter near the Sun. His “Kapteyn Universe” is the first time dark matter is treated as a quantity to be measured rather than a possibility to be entertained.

6. 1922 — Jeans sharpens it. Reanalyzing Kapteyn’s data, James Jeans concluded there might be roughly two dark stars for every bright one — the first claim that the dark component *dominates*.

7. 1932 — Oort, and the first false alarm. Jan Oort measured the vertical oscillations of nearby stars and inferred that the mass in the galactic disk exceeds the visible mass — the “Oort limit.” It is important to record what happened to this result: it was *wrong*. Better data eventually showed the local disk density is consistent with the stars and gas we see, with little or no room for dark matter. The first quantitative detection of galactic dark matter was a measurement error. This is not a slander on Oort; it is a reminder that “the gravity doesn’t add up” has, from the very beginning, sometimes meant “the bookkeeping was off.”

PART III — The cluster and galaxy evidence (1933–1972)

8. 1933 — Zwicky and the missing mass of Coma. Fritz Zwicky applied the virial theorem to the Coma cluster of galaxies: the galaxies were moving far too fast to be bound by the gravity of their visible mass. He inferred a large amount of *dunkle Materie* — coining the German phrase — and estimated the missing mass at a startling factor (hundreds, on the distance scale of his day; smaller but still dominant on modern scales). This is the first robust, hard-to-dismiss case, and the true birth of the modern problem. It was then largely ignored for forty years.

9. 1936 — Smith confirms it in Virgo. Sinclair Smith found the same mass discrepancy in the Virgo cluster. Two clusters, same verdict.

10. 1939 — Babcock sees it in a galaxy. Horace Babcock measured the rotation of the Andromeda galaxy (M31) and found the outer regions rotating faster than the visible mass should allow — the first hint of the flat rotation curve, though attributed at the time to other effects.

11. 1940s–1950s — dormancy. The “missing mass problem” in clusters was known and occasionally discussed (the term itself was used), but dark matter was a curiosity at the edge of the field, not a research program.

12. 1959 — the Local Group weighs in. Franz Kahn and Lodewijk Woltjer used the fact that Andromeda is *falling toward* the Milky Way (rather than receding with the expansion) to argue, via a timing argument, for substantial unseen mass binding the Local Group.

PART IV — The 1970s revolution: dark matter becomes real (1970–1980)

13. 1970 — Rubin and Ford reopen the case. Vera Rubin and Kent Ford measured the rotation of Andromeda with new precision and found the curve *flat* far out — stars at the edge circling as fast as those nearer in, exactly what should not happen if the mass follows the light.

14. 1970 — Freeman notices the gap. Ken Freeman, studying galaxy disks, remarked that if the rotation curves were as flat as they appeared, there must be additional mass not following the light.

15. 1970s — flat curves everywhere. Through the decade Rubin, Ford, and Thonnard measured dozens of spirals; Morton Roberts and, decisively, Albert Bosma (1978) used radio observations of hydrogen gas to trace rotation far beyond the visible disk. The flat rotation curve went from oddity to universal rule. Every spiral galaxy seemed to demand a massive invisible halo.

16. 1973–74 — the theoretical clincher and the consensus. Jerry Ostriker and Jim Peebles showed (1973) that a galactic disk *without* a massive surrounding halo would be dynamically unstable and tear itself apart; halos were needed for galaxies to exist at all. In 1974 two independent papers — Ostriker, Peebles & Yahil in the United States, and Einasto, Kaasik & Saar in Estonia — assembled the evidence and concluded that galaxies are embedded in halos roughly *ten times* their visible mass. This is the moment, around 1974, when dark matter became mainstream astrophysics. **The model’s first firm form: galaxies and clusters sit in extended halos of unseen, gravitating matter, several times the visible mass.**

PART V — What is it? The candidate wars (1977–1984)

17. Late 1970s — the first great constraint: it can’t (mostly) be ordinary matter. As Big Bang nucleosynthesis calculations matured, they pinned the total amount of *baryonic* (ordinary) matter — protons and neutrons — to a value far below the total mass the halos required. If the nucleosynthesis was right, most of the dark matter could not be made of normal atoms, dust, dead stars, or gas. **This forced the first deep change: dark matter became, by elimination, a new kind of particle not yet in the Standard Model of physics.** The problem migrated from astronomy into particle physics.

18. 1977 — the candidates are born. Two ideas arrived almost together. The axion was proposed (Peccei–Quinn; Weinberg; Wilczek) to fix an unrelated problem in the strong nuclear force, and turned out to make a fine dark-matter candidate. And Lee and Weinberg worked out the relic abundance of a heavy, weakly interacting particle — the foundation of the **WIMP** (Weakly Interacting Massive Particle), whose “miracle” was that a particle at the weak-force scale would naturally freeze out with about the right density. For the next forty years the WIMP would be the front-runner.

19. 1980 — hot dark matter’s brief reign. A Soviet experiment (Lyubimov) claimed the neutrino had a mass around 30 electron-volts — enough for the universe’s neutrinos to *be* the dark matter. Neutrinos were known to exist, so this was the economical answer, and for a few years “hot dark matter” was the favorite.

20. 1982–83 — hot dark matter falls. Light, fast neutrinos stream out of small clumps before those clumps can collapse, so hot dark matter builds the largest structures first and lets galaxies fragment out of them later — a “top-down” universe. The galaxy surveys showed the opposite: small structures formed first, “bottom-up.” Simulations (White, Frenk & Davis, 1983) confirmed hot dark matter produced the wrong universe. **The candidate was changed again — this time by its temperature.**

21. 1982–84 — the cold dark matter revolution. Peebles (1982) and then Blumenthal, Faber, Primack & Rees (1984) developed **cold dark matter (CDM)**: heavy, slow particles that clump on every scale and build structure bottom-up. CDM fit the growth of galaxies and large-scale structure far better than hot dark matter. **This is the central particle model that survives, in modified form, to today: the “CDM” in Λ CDM.** Notice the pattern across steps 17–21: confronted with conflict, the field did not revisit gravity; it re-specified the invisible particle — first making it non-baryonic, then cold instead of hot.

22. 1983 — the alternative is born. In the same window, Mordehai Milgrom proposed **Modified Newtonian Dynamics (MOND)**: that below a tiny, universal acceleration scale, the law of motion itself changes, removing the need for halos in galaxies. This is the Vulcan branch reasserting itself — *missing physics, not missing matter* — and it would shadow the dark-matter program ever after, fitting galaxy rotation curves with uncanny economy while struggling with clusters and cosmology.

PART VI — Standard CDM and the long road to Λ (1984–1998)

23. mid-1980s — Standard CDM becomes the default. Inflation predicted a spatially flat universe, which (assuming matter supplies the flatness) means the total matter density equals the critical density: $\Omega_m = 1$. Combined with cold dark matter, this gave **Standard CDM (SCDM)**, the establishment model for over a decade.

24. 1990 — the first warning that Λ was needed. Efstathiou, Sutherland & Maddox analyzed the clustering of galaxies in the APM survey and argued, in print, that the data fit poorly unless the matter density was *low* ($\Omega_m \approx 0.3$) and the rest of the critical density was supplied by a **cosmological constant Λ** . This is eight years before the supernovae. The signal for dark energy was already visible in the galaxy distribution — and was largely set aside, because adding Λ was unfashionable.

25. 1992 — COBE tightens the screws. The COBE satellite detected the tiny temperature ripples in the cosmic microwave background. Normalizing SCDM to those ripples made it over-produce clumpiness on small scales: SCDM predicted a universe lumpier than the one we live in.

26. early–mid 1990s — SCDM in crisis. Several independent failures stacked up: the **age problem** (an $\Omega_m = 1$ universe came out younger than its oldest stars); the **baryon catastrophe**

(White, Navarro, Evrard & Frenk, 1993, showed galaxy clusters hold too high a fraction of ordinary matter for $\Omega_m = 1$); and the **shape problem** (galaxy clustering implied less matter than SCDM). **The response was not one fix but a market of them:** tilted CDM, mixed cold-plus-hot CDM (CHDM), open low-density CDM (OCDM), decaying-particle τ CDM, and Λ CDM as merely one option among many. A decade of model proliferation — the classic symptom of a paradigm under strain.

27. 1998–99 — dark energy, and the model crystallizes. Two teams measuring distant Type Ia supernovae (Riess and collaborators, 1998; Perlmutter and collaborators, 1999) found the cosmic expansion *accelerating*. That demanded a component with negative pressure — the cosmological constant Λ . Lowering matter to $\Omega_m \approx 0.3$ and letting Λ supply $\Omega_\Lambda \approx 0.7$ resolved the age problem, the baryon fraction, and the clustering shape *simultaneously*, and matched the new supernova data. **SCDM became Λ CDM: ~5% ordinary matter, ~27% cold dark matter, ~68% dark energy.** The model now required two unseen components, together ~95% of everything — and the value of Λ remained (and remains) unexplained, missing the natural quantum-field-theory estimate by some 120 orders of magnitude.

PART VII — Concordance: the genuine triumph (2000–2010)

28. 2003 — the prediction that earned the crown. The WMAP satellite mapped the acoustic peaks of the cosmic microwave background. Λ CDM predicts the entire pattern of those peaks from six numbers, and they *matched*, with precision. This was a real, risky, confirmed prediction — not an accommodation — and it is the strongest single reason Λ CDM is the standard model. An honest history credits it without reservation.

29. 2005 — the cosmic ruler confirmed. The same six-parameter model predicted a specific length scale — the baryon acoustic oscillation “ruler” — imprinted in the spacing of galaxies. The Sloan Digital Sky Survey and the 2dF survey found it, right where predicted. A second clean win.

30. 2006 — the Bullet Cluster: dark matter’s best evidence. Clowe and collaborators imaged two colliding galaxy clusters. The hot gas (most of the ordinary matter) was slowed by the collision and lagged behind, while gravitational lensing showed most of the *mass* sailing on ahead with the galaxies — as if dominated by something that barely interacts. This is widely regarded as the most direct empirical case for collisionless dark matter, and a serious difficulty for the simplest modified-gravity theories. It belongs in any honest account, and it cuts against the alternative tradition. (Modified-inertia and relativistic MOND-type theories have since offered partial responses, and clusters remain a genuine difficulty for *that* side too.)

PART VIII — The small-scale crisis and the proliferating zoo (1994–2016)

31. 1994–1999 — CDM fails inside galaxies. As simulations resolved smaller scales, cold dark matter began to mispredict the details: - **The cusp–core problem** (Flores & Primack; Moore, 1994): CDM halos should rise to a dense central “cusp,” but many dwarf galaxies show flat central “cores.” - **The missing-satellites problem** (Klypin et al.; Moore et al., 1999): CDM predicts

hundreds to thousands of small subhalos around a galaxy like ours; only dozens of satellites are seen.

32. 2000 — the particle gets modified again. Spergel & Steinhardt proposed **self-interacting dark matter (SIDM)** — letting the dark particle bump into itself — specifically to turn cusps into cores. A new parameter (the self-interaction strength) added to fix a specific failure.

33. 2000s — MACHOs eliminated. Long microlensing campaigns (MACHO, EROS, OGLE) searched for dark *baryonic* lumps — faint stars, planets, black holes — passing in front of background stars. They found far too few. **Compact ordinary objects were ruled out as the bulk of the dark matter**, reinforcing that it must be exotic.

34. 2011 — “too big to fail.” Boylan-Kolchin, Bullock & Kaplinghat showed the biggest predicted subhalos are too dense to match the observed satellites: they are “too big to fail” to form visible galaxies, yet apparently did. Another mismatch between CDM’s predictions and the local universe.

35. 2000s–2010s — the candidate zoo expands. With the WIMP not showing up and the small-scale problems persisting, the menu of dark-matter particles grew: **warm dark matter** (a keV-scale sterile neutrino, to erase excess small structure), **fuzzy / ultralight dark matter** (an axion-like particle of mass $\sim 10^{-22}$ eV, whose quantum wavelength is galaxy-sized, to make cores), and a revival of **primordial black holes** as dark matter (re-energized in 2016 after LIGO’s black-hole detections). Each is a different invisible thing, each tuned to relieve a different tension.

36. mid-2010s — the simulators answer with feedback. The mainstream response to the small-scale crisis was not a new particle but new *astrophysics*: large simulations (FIRE, EAGLE, IllustrisTNG) added supernova winds and black-hole outflows — “baryonic feedback” — tuned until cusps softened, satellites dimmed, and the counts came out right. The failures were absorbed by calibrating processes inside the simulations to reproduce the very data they were meant to explain.

37. 2016 — the relation dark matter must conspire to fit. McGaugh, Lelli & Schombert published the **radial acceleration relation**: across 153 galaxies, the observed gravity is fixed almost entirely by the visible matter, through a single universal acceleration scale appearing in every galaxy. This is natural in modified-gravity theories (it is what they predict) and a fine-tuning puzzle for dark matter, which must arrange its invisible halos, galaxy by galaxy, to reproduce a law it has no reason to know.

PART IX — The non-detection (1986–2024)

38. 1986–2010s — the great hunt begins. Direct-detection experiments (after Goodman & Witten and Drukier, Freese & Spergel laid the groundwork) went underground to catch a dark particle recoiling off a nucleus: DAMA (whose disputed annual-modulation signal was never reproduced by others), then CDMS, XENON, LUX, PandaX.

39. 2017–2023 — the WIMP is cornered. Ever-larger detectors — XENON1T, LUX-ZEPLIN (LZ, 2022–23), PandaX-4T — found *nothing*, pushing the limits down toward the “neutrino floor,”

the point where neutrinos themselves become an irreducible background. In parallel, the Large Hadron Collider found no supersymmetry and no WIMPs. **The favored candidate of forty years has been excluded across most of its natural parameter space** without ever appearing. The model's response has been to shift attention to other candidates (axions, ultralight, primordial black holes) rather than to revisit the premise.

40. 2009–2020s — indirect-detection false dawns. Hints that looked like dark matter annihilating — the galactic-center gamma-ray excess, a 3.5 keV X-ray line — each generated excitement and then faded into more mundane astrophysical explanations or unconfirmed status. No clean indirect signal has survived.

PART X — The tensions era (2013–present)

41. 2013–2018 — Planck perfects the concordance, and exposes the cracks. The Planck satellite measured the CMB to exquisite precision, nailing the six Λ CDM parameters — and in doing so sharpened two disagreements with other data.

42. ~2016–present — the Hubble tension. The present expansion rate inferred from the CMB ($H_0 \approx 67.4$) disagrees with the rate measured directly from nearby supernovae and Cepheids ($H_0 \approx 73$) at more than **five standard deviations**. Proposed fixes are, characteristically, additions: **early dark energy** switched on just before recombination, extra relativistic species, or modified recombination physics.

43. ~2017–present — the S_8 tension. Gravitational-lensing and cluster surveys (KiDS, DES) find the universe slightly *less* clumpy than CMB-calibrated Λ CDM predicts — a two-to-three-sigma mismatch in how fast structure has grown.

44. 2022–present — JWST's early galaxies. The James Webb Space Telescope found galaxies at very high redshift looking more massive and mature than Λ CDM's structure-formation timetable comfortably allows — a tension still being argued over, with the usual menu of fixes (revised star formation, new physics) on the table.

45. 2024–2025 — DESI and the un-fixing of Λ . The Dark Energy Spectroscopic Instrument's baryon-acoustic measurements *prefer a dark energy that changes with time* ($w \neq -1$) over a constant Λ , at a significance climbing toward four sigma in some dataset combinations. **If this holds, the one component Λ CDM treats as a fixed constant is not constant** — and the model must absorb a dynamical dark sector ($w_0 w_a$ CDM), trading one fixed number for two free functions. The cosmological-constant problem is not solved by this; it is relocated.

PART XI — The cosmic dipole: a challenge to the foundation (2021–present)

46. The assumption underneath everything. Λ CDM rests on the **cosmological principle**: on large scales the universe is the same in every direction (isotropic) and at every place (homogeneous). This is less a measured fact than the simplifying assumption that makes the entire mathematical framework (Friedmann–Lemaître–Robertson–Walker) work. Almost everything in cosmology is built on it.

47. The clean test. Our motion through the universe should make the distant sky look slightly more crowded ahead of us and sparser behind — a “dipole.” The cosmic microwave background shows exactly such a dipole, giving our speed as about 370 km/s. The cosmological principle then makes a sharp, freedom-free prediction: the dipole counted in *any* distant population, from our motion alone, must match the CMB’s in direction and amplitude.

48. 2021 — Secrest and collaborators break the prediction. Using 1.36 million quasars from the space-based WISE survey — chosen precisely to avoid the ground-based systematics that haunt radio catalogs — Nathan Secrest and colleagues measured the quasar dipole. It pointed roughly the right way, but its amplitude was **more than twice** what our motion predicts, rejecting the purely-kinematic interpretation at a p-value near 5×10^{-7} — **4.9 σ** .

49. 2022 — independent data, same verdict. Combining the WISE quasars with NVSS radio galaxies — a fully independent dataset — Secrest and collaborators tightened the result to **5.1 σ** and titled the paper, without hedging, *A Challenge to the Standard Cosmological Model*. Read plainly: either we move about twice as fast as the CMB says (which conflicts with the CMB itself), or the distant universe is genuinely anisotropic and the cosmological principle is wrong at a level that matters. Unlike the H_0 or S_8 tensions, this cannot be patched by adjusting how much dark matter or dark energy there is. It questions the symmetry the whole model is built on.

50. 2023–2025 — the anomaly is contested, not settled. Honesty requires the other half. New analyses — Bayesian dipole estimators, reworked CatWISE quasar samples, careful accounting of the kinematic contribution to source counts and of flux- and redshift-evolution effects — are actively testing whether part or all of the excess is an unmodeled systematic rather than a true cosmic anisotropy. The dipole is a sharp, foundational, persistent anomaly whose verdict is still being argued. That is exactly the kind of measurement to watch, and exactly the kind the model has not yet shown it can absorb.

PART XII — The pattern, and the adjustment to come

Step back and read the sequence as a whole. The *model* changed at least a dozen times, and almost every change was a response to data the model had not predicted:

- the dark matter became **non-baryonic** (forced by nucleosynthesis, step 17);
- it became **cold instead of hot** (forced by structure, step 21);
- the universe gained a whole new component, **dark energy** (forced by supernovae, step 27);
- the small-scale failures were met with **feedback tuning** and a **growing zoo of candidate particles** (steps 31–36);
- the never-detected WIMP was quietly **demoted in favor of other candidates** rather than abandoned (step 39);
- and the present tensions are being met with **early dark energy**, **evolving w** , and — if the dipole survives — some controlled **breaking of isotropy** (steps 42–50).

On this track record, the next move is easy to forecast in *shape*. Λ CDM will most likely be adjusted again, in one of its three customary ways:

1. **Promote a constant to a function** — already underway, as DESI pushes a fixed Λ toward a time-varying $w(z)$, adding parameters where there was one number.
2. **Add a component that switches on at the right moment** — the early-dark-energy template, engineered to close the Hubble gap and otherwise leave the successes intact.
3. **Weaken a foundational assumption, locally and just enough** — what the dipole will demand if it holds: a “tilted” universe, a very-large-scale structure, or an anisotropic mode, added with enough freedom to fit the observed dipole while preserving the CMB’s near-perfect isotropy.

In every case the model survives by gaining a part. This is not a proof that Λ CDM is false — extended models may be true, and refining a theory against better data is sometimes simply how science works. It is a statement about the model’s *mode*: for ninety years it has met anomalies by absorbing them, and the safe bet is that it will do so again.

PART XIII — The crux, and the other road

Philosophers of science distinguish a *progressive* research program — one that absorbs anomalies while making new, confirmed predictions — from a *degenerative* one, which absorbs them while predicting nothing new, each fix bought only to cover a known failure. The Ptolemaic astronomers who stacked epicycle upon epicycle were not foolish, and their model fit the data beautifully to the very end. It was simply doing far more accommodating than predicting, while a simpler picture waited to be seen.

Λ CDM is honestly *both*. It has a spine of genuine, risky, confirmed predictions — the CMB acoustic peaks, the acoustic ruler, the Bullet Cluster’s collisionless mass — that no alternative yet matches. And it has a long, lengthening tail of after-the-fact accommodations: the temperature of the dark matter, the addition of Λ , the feedback parameters, the candidate zoo, early dark energy, evolving w , and perhaps soon a broken symmetry. Which list grows faster in the decade now opening is the entire question.

It is worth at least noticing that the rival tradition — the Vulcan branch, modified dynamics — behaves differently. The emergent-gravity proposal that ties the single galactic acceleration scale to the cosmological constant, $a_0 = c^2 \sqrt{(\Lambda/32\pi)}$, earns its keep by *prediction*: a fixed number for the scale, a falsifiable redshift trend (a declining $a_0(z)$), a solar-system test (Cassini) it has already either passed or failed. Such theories are far less developed than Λ CDM, carry real unsolved problems of their own — clusters, the Bullet Cluster, the unbuilt covariant completion — and may well be wrong. But they share a virtue the standard model has slowly traded away across the fifty steps above: they can be *killed* by a single measurement, because they were not built with the freedom to absorb it.

That is the lesson of the long chronology. The dark-matter model began, in 1933, as a bold and reasonable inference, and it grew into the most accurate cosmological framework ever written. But somewhere along the way it also became something else — a structure flexible enough to fit Neptune and Vulcan alike, to call any new datum either a confirmation or a parameter. The cosmic dipole is the latest test of whether there is any measurement it cannot fit. If one finally arrives that refuses to be absorbed, history suggests it will be answered, in the end, not by the

model that learned to fit everything, but by one that was brave enough to name its number in advance.

Sources and further reading

A standard scholarly account is Bertone & Hooper, *A History of Dark Matter*, Reviews of Modern Physics **90**, 045002 (2018) [arXiv:1605.04909].

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- Secrest et al. (2021), *ApJL* **908**, L51; (2022), *ApJL* **937**, L31 — the cosmic dipole; and the 2023–2025 reassessment literature (Bayesian estimators; CatWISE re-analyses; kinematic number-count contributions).

An interpretive chronology. Λ CDM’s genuine predictive successes (the CMB peaks, the acoustic ruler, the Bullet Cluster) are credited in full; the dark-matter model’s repeated post-hoc revisions are documented step by step; the cosmic dipole is presented as a sharp but still-contested anomaly, not a settled refutation. The forecast concerns the model’s mode of response, not its truth. The author’s own emergent-gravity proposal is noted as an alternative, with its unsolved problems (clusters, the Bullet Cluster, the unbuilt covariant completion) acknowledged. Related framework DOI: 10.5281/zenodo.20721540.